

Example of Constructing a Predictive Parsing Table

****FIRST(α):**** Set of terminals that appear first in strings derived from α (and includes ϵ if $\alpha \Rightarrow^* \epsilon$).

****FOLLOW(A):**** Set of terminals that can appear immediately after non-terminal A in some sentential form.

Grammar:

$$\begin{array}{lll}
 \underline{e} & \rightarrow & \underline{t} \underline{e}' \\
 \underline{e}' & \rightarrow & + \underline{t} \underline{e}' \\
 & / & \varepsilon \\
 \underline{t} & \rightarrow & \underline{f} \underline{t}' \\
 \underline{t}' & \rightarrow & * \underline{f} \underline{t}' \\
 & / & \varepsilon \\
 \underline{f} & \rightarrow & (\underline{e}) \\
 & / & x \\
 & / & y
 \end{array}$$

first and ***follow*** sets:

Non-Terminal	<i>first</i>	<i>follow</i>
$\underline{e}$	'(', 'x', 'y'	\$, ')'
$\underline{e}'$	+', ε	\$, ')'
$\underline{t}$	'(', 'x', 'y'	+', \$, ')'
$\underline{t}'$	*, ε	+', \$, ')'
$\underline{f}$	'(', 'x', 'y'	*, '+', ')', \$

Start with an empty parsing table; the **rows** are non-terminals and the **columns** are terminals.

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$							
$\underline{e'}$							
$\underline{t}$							
$\underline{t'}$							
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
if $\varepsilon \in \text{first}(\alpha)$ then
for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{e} \rightarrow \underline{t} \underline{e}'$

$\text{first}(\underline{t} \underline{e}') = '(', 'x', 'y'$

Non-Terminal	Input Symbol						
	x	y	+	*	(	)	\$
$\underline{e}$					$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$							
$\underline{t}$							
$\underline{t}'$							
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
if $\varepsilon \in \text{first}(\alpha)$ then
for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{e} \rightarrow \underline{t} \underline{e}'$

$\text{first}(\underline{t} \underline{e}') = '(', \boxed{'x'}, 'y'$

Non-Terminal	Input Symbol						
	x	y	+	*	(	)	\$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e}'$				$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$							
$\underline{t}$							
$\underline{t}'$							
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{e} \rightarrow \underline{t} \underline{e}'$

$\text{first}(\underline{t} \underline{e}') = '(', 'x', 'y'$

Non-Terminal	Input Symbol						
	x	y	+	*	(	)	\$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e}'$	$\underline{e} \rightarrow \underline{t} \underline{e}'$			$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$							
$\underline{t}$							
$\underline{t}'$							
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{e'} \rightarrow + \underline{t} \underline{e'}$

$\text{first}(+ \underline{t} \underline{e'}) = '+'$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$			$\underline{e} \rightarrow \underline{t} \underline{e'}$		
$\underline{e'}$			$\underline{e'} \rightarrow + \underline{t} \underline{e'}$				
$\underline{t}$							
$\underline{t'}$							
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{e'} \rightarrow \varepsilon$

$\text{follow}(\underline{e'}) = \$, ')'$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$			$\underline{e} \rightarrow \underline{t} \underline{e'}$		
$\underline{e'}$			$\underline{e'} \rightarrow + \underline{t} \underline{e'}$				$\underline{e'} \rightarrow \varepsilon$
$\underline{t}$							
$\underline{t'}$							
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{e'} \rightarrow \varepsilon$

$\text{follow}(\underline{e'}) = \$, \text{'})'$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$			$\underline{e} \rightarrow \underline{t} \underline{e'}$		
$\underline{e'}$			$\underline{e'} \rightarrow + \underline{t} \underline{e'}$			$\underline{e'} \rightarrow \varepsilon$	$\underline{e'} \rightarrow \varepsilon$
$\underline{t}$							
$\underline{t'}$							
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{t} \rightarrow f \underline{t}'$

$\text{first}(f \underline{t}') = '(', 'x', 'y'$

Non-Terminal	Input Symbol						
	x	y	+	*	(	)	\$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e}'$	$\underline{e} \rightarrow \underline{t} \underline{e}'$			$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$			$\underline{e}' \rightarrow + \underline{t} \underline{e}'$			$\underline{e}' \rightarrow \varepsilon$	$\underline{e}' \rightarrow \varepsilon$
$\underline{t}$					$\underline{t} \rightarrow f \underline{t}'$		
$\underline{t}'$							
f							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{t} \rightarrow f \underline{t}'$

$\text{first}(f \underline{t}') = '(', 'x', 'y'$

Non-Terminal	Input Symbol						
	x	y	+	*	(	)	\$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e}'$	$\underline{e} \rightarrow \underline{t} \underline{e}'$			$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$			$\underline{e}' \rightarrow + \underline{t} \underline{e}'$			$\underline{e}' \rightarrow \varepsilon$	$\underline{e}' \rightarrow \varepsilon$
$\underline{t}$	$\underline{t} \rightarrow f \underline{t}'$				$\underline{t} \rightarrow f \underline{t}'$		
$\underline{t}'$							
f							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{t} \rightarrow f \underline{t}'$

$\text{first}(f \underline{t}') = '(', 'x', \textbf{'y'}$

Non-Terminal	Input Symbol						
	x	y	+	*	(	)	\$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e}'$	$\underline{e} \rightarrow \underline{t} \underline{e}'$			$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$			$\underline{e}' \rightarrow + \underline{t} \underline{e}'$			$\underline{e}' \rightarrow \varepsilon$	$\underline{e}' \rightarrow \varepsilon$
$\underline{t}$	$\underline{t} \rightarrow f \underline{t}'$	$\underline{t} \rightarrow f \underline{t}'$			$\underline{t} \rightarrow f \underline{t}'$		
$\underline{t}'$							
f							

for each production $\underline{n} \rightarrow \alpha$
for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
if $\varepsilon \in \text{first}(\alpha)$ then
for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{t'} \rightarrow \alpha$

$\text{follow}(\underline{t'}) = \text{'+'}, \$, \text{'})'$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$			$\underline{e} \rightarrow \underline{t} \underline{e'}$		
$\underline{e'}$			$\underline{e'} \rightarrow + \underline{t} \underline{e'}$			$\underline{e'} \rightarrow \varepsilon$	$\underline{e'} \rightarrow \varepsilon$
$\underline{t}$	$\underline{t} \rightarrow \underline{f} \underline{t'}$	$\underline{t} \rightarrow \underline{f} \underline{t'}$			$\underline{t} \rightarrow \underline{f} \underline{t'}$		
$\underline{t'}$			$\underline{t'} \rightarrow \varepsilon$	$\underline{t'} \rightarrow * \underline{f} \underline{t'}$			
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{t'} \rightarrow \alpha$

$\text{follow}(\underline{t'}) = \text{'+'}, \text{'$'}, \text{'('}$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$			$\underline{e} \rightarrow \underline{t} \underline{e'}$		
$\underline{e'}$			$\underline{e'} \rightarrow + \underline{t} \underline{e'}$			$\underline{e'} \rightarrow \varepsilon$	$\underline{e'} \rightarrow \varepsilon$
$\underline{t}$	$\underline{t} \rightarrow \underline{f} \underline{t'}$	$\underline{t} \rightarrow \underline{f} \underline{t'}$			$\underline{t} \rightarrow \underline{f} \underline{t'}$		
$\underline{t'}$			$\underline{t'} \rightarrow \varepsilon$	$\underline{t'} \rightarrow * \underline{f} \underline{t'}$			$\underline{t'} \rightarrow \varepsilon$
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{t'} \rightarrow \alpha$

$\text{follow}(\underline{t'}) = \text{'+'}, \$, \text{'(')}$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$	$\underline{e} \rightarrow \underline{t} \underline{e'}$			$\underline{e} \rightarrow \underline{t} \underline{e'}$		
$\underline{e'}$			$\underline{e'} \rightarrow + \underline{t} \underline{e'}$			$\underline{e'} \rightarrow \varepsilon$	$\underline{e'} \rightarrow \varepsilon$
$\underline{t}$	$\underline{t} \rightarrow \underline{f} \underline{t'}$	$\underline{t} \rightarrow \underline{f} \underline{t'}$			$\underline{t} \rightarrow \underline{f} \underline{t'}$		
$\underline{t'}$			$\underline{t'} \rightarrow \varepsilon$	$\underline{t'} \rightarrow * \underline{f} \underline{t'}$		$\underline{t'} \rightarrow \varepsilon$	$\underline{t'} \rightarrow \varepsilon$
$\underline{f}$							

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{f} \rightarrow (\underline{e})$

$\text{first}((\underline{e})) = '('$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e}'$	$\underline{e} \rightarrow \underline{t} \underline{e}'$			$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$			$\underline{e}' \rightarrow + \underline{t} \underline{e}'$			$\underline{e}' \rightarrow \varepsilon$	$\underline{e}' \rightarrow \varepsilon$
$\underline{t}$	$\underline{t} \rightarrow \underline{f} \underline{t}'$	$\underline{t} \rightarrow \underline{f} \underline{t}'$			$\underline{t} \rightarrow \underline{f} \underline{t}'$		
$\underline{t}'$			$\underline{t}' \rightarrow \varepsilon$	$\underline{t}' \rightarrow * \underline{f} \underline{t}'$		$\underline{t}' \rightarrow \varepsilon$	$\underline{t}' \rightarrow \varepsilon$
$\underline{f}$					$\underline{f} \rightarrow (\underline{e})$		

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{f} \rightarrow x$

$\text{first}(x) = 'x'$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e}'$	$\underline{e} \rightarrow \underline{t} \underline{e}'$			$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$			$\underline{e}' \rightarrow + \underline{t} \underline{e}'$			$\underline{e}' \rightarrow \varepsilon$	$\underline{e}' \rightarrow \varepsilon$
$\underline{t}$	$\underline{t} \rightarrow \underline{f} \underline{t}'$	$\underline{t} \rightarrow \underline{f} \underline{t}'$			$\underline{t} \rightarrow \underline{f} \underline{t}'$		
$\underline{t}'$			$\underline{t}' \rightarrow \varepsilon$	$\underline{t}' \rightarrow * \underline{f} \underline{t}'$		$\underline{t}' \rightarrow \varepsilon$	$\underline{t}' \rightarrow \varepsilon$
$\underline{f}$	$\underline{f} \rightarrow x$				$\underline{f} \rightarrow (\underline{e})$		

for each production $\underline{n} \rightarrow \alpha$
 for each $a \in \text{first}(\alpha)$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$
 if $\varepsilon \in \text{first}(\alpha)$ then
 for each $b \in \text{follow}(\underline{n})$
 add $\underline{n} \rightarrow \alpha$ to $T[\underline{n}, a]$

Current production:

$\underline{f} \rightarrow y$

$\text{first}(y) = 'y'$

Non-Terminal	Input Symbol						
	x	y	$+$	$*$	$($	$)$	$\$$
$\underline{e}$	$\underline{e} \rightarrow \underline{t} \underline{e}'$	$\underline{e} \rightarrow \underline{t} \underline{e}'$			$\underline{e} \rightarrow \underline{t} \underline{e}'$		
$\underline{e}'$			$\underline{e}' \rightarrow + \underline{t} \underline{e}'$			$\underline{e}' \rightarrow \varepsilon$	$\underline{e}' \rightarrow \varepsilon$
$\underline{t}$	$\underline{t} \rightarrow \underline{f} \underline{t}'$	$\underline{t} \rightarrow \underline{f} \underline{t}'$			$\underline{t} \rightarrow \underline{f} \underline{t}'$		
$\underline{t}'$			$\underline{t}' \rightarrow \varepsilon$	$\underline{t}' \rightarrow * \underline{f} \underline{t}'$		$\underline{t}' \rightarrow \varepsilon$	$\underline{t}' \rightarrow \varepsilon$
$\underline{f}$	$\underline{f} \rightarrow x$	$\underline{f} \rightarrow y$			$\underline{f} \rightarrow (\underline{e})$		